

EE1060: Differential Equations and Transform Techniques

Practice Problem Set - 01

Practice problems in topics covered till: Lecture 04

Note : The practice problem sets are designed for your personal review and are not required for submission or evaluation. These problems aim to reinforce your understanding of the material and help you prepare effectively. Some of the practice problems are sourced from the course references, and it is strongly recommended that you consult these references for additional practice problems. Due credit is also given to the TAs of the course, who have contributed to the creation of this problem set.

1. Determine whether the following differential equations are linear and identify their order.

(a) $x^2 \frac{d^2 y}{dx^2} + \frac{dy}{dx} + y = 10x^3$

(b) $x^2 y \frac{dy}{dx} - 10x^3 = 0$

(c) $x^2 \left(\frac{d^2 y}{dx^2} \right)^2 + \frac{dy}{dx} + y = 0$

(d) $x^2 \left(\frac{d^2 y}{dx^2} \right)^2 + (3x + 4) \frac{dy}{dx} + y^2 = 0$

(e) $\sin x \frac{d^2 y}{dx^2} + \frac{dy}{dx} - xy = 0$

(f) $\frac{d^2 y}{dx^2} - \beta(1 - y^2) \frac{dy}{dx} + y = 0$

(g) $\frac{\partial y}{\partial t} + y \frac{\partial y}{\partial x} - \alpha \frac{\partial^2 y}{\partial x^2} = 0$, where α and β are constants.

(h) $\frac{d^2 y}{dx^2} + \alpha \frac{dy}{dx} + \beta y + ay^3 = A \cos \omega x$.

(i) $\frac{dy}{dx} - A(x) - B(x)y - c(x)y^2 = 0$.

2. Show that $y = cx - x^2$ is a solution to the ODE. Also find the particular solution for the initial condition $y(1) = 2$.

$$x \frac{dy}{dx} - y + x^2 = 0 \quad (1)$$

3. Show that $y = c_1 e^{-at} + c_2 e^{-bt}$ forms the 2-parameter solution to the differential equation

$$\frac{d^2 y}{dx^2} + (a + b) \frac{dy}{dx} + aby = 0 \quad (5)$$

4. Compute the current response of a series LC circuit with the initial condition $v_c(0) = V_0$, where v_c is the voltage across the capacitor. For reference, the current response of a series LC circuit is governed by the DE

$$\frac{d^2 i}{dt^2} + \frac{1}{LC} i(t) = 0 \quad (9)$$

5. Determine the value of α for which the current $i(t) = Ae^{-\alpha t} \cos(\omega t + \phi)$ where $\omega = \sqrt{\frac{1}{LC} - \alpha^2}$, forms the general solution to the second-order linear differential equation

$$L \frac{d^2 i}{dt^2} + R \frac{di}{dt} + \frac{1}{C} i = 0, \quad (13)$$

6. Determine the value of β for which $y(x) = e^{\beta x}$ is a solution to the differential equation

$$\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = 0 \quad (18)$$

7. Determine the value of n for which $y(x) = 2(x^n - 1) + ce^{-(x^n)}$ is a solution to the differential equation

$$\frac{dy}{dx} + 2xy = 4x^3 \quad (21)$$

8. Solve the first-order ODE

$$\frac{dy}{dx} = y^2 e^x \quad (24)$$

9. Compute the particular solution for the first-order ODE

$$(y^2 + 1) \frac{dy}{dx} = x \text{ with the initial condition } y(0) = 1 \quad (27)$$